Gravitational Lensing in the Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)

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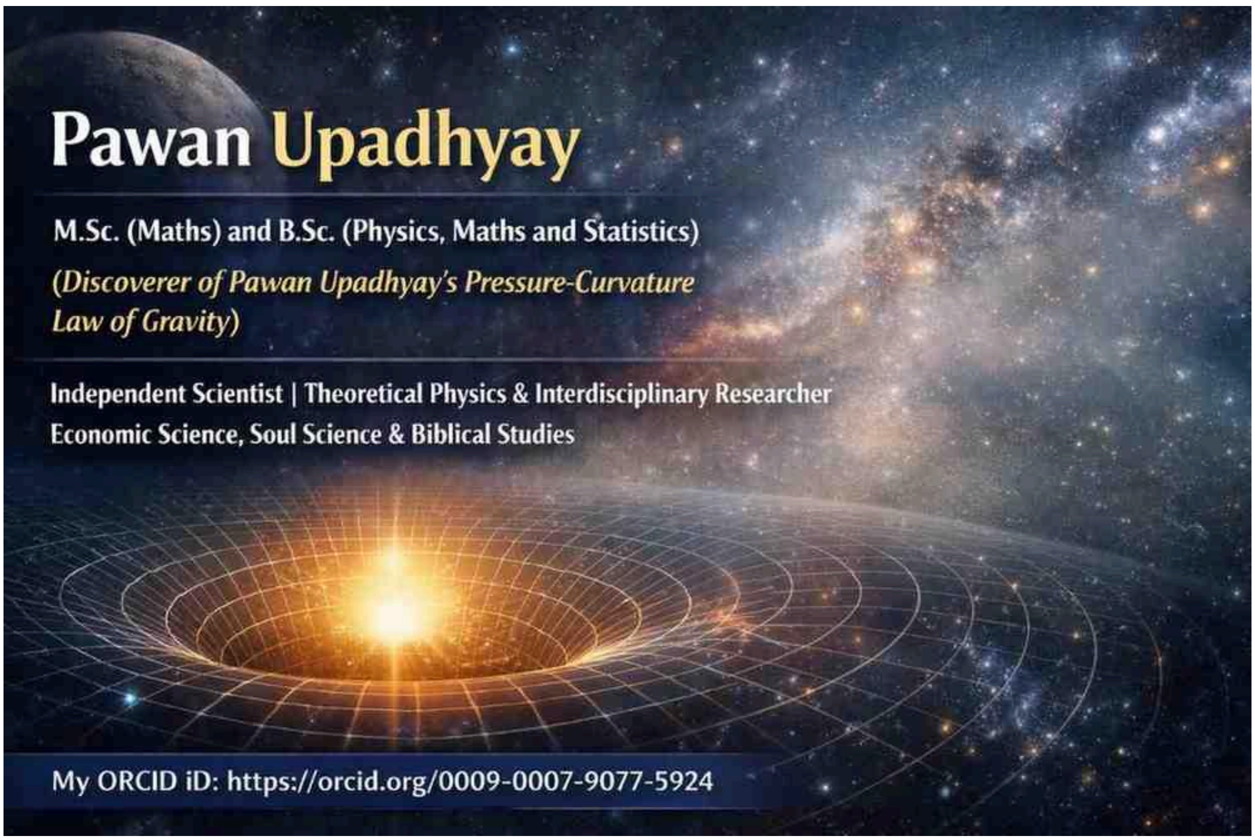

Abstract

We present a consistent weak-field formulation of gravitational lensing within the Pressure–Potential Curvature (PPC) law of gravity. In this framework, gravity is sourced by a pressure-augmented energy density, with the effective source for a perfect (isotropic) fluid given by (E_d + 3P_g). We derive the PPC Poisson equation, the light-deflection law, and the angular lensing relations, including the Einstein radius. The formulation reproduces the standard weak-field structure of relativistic gravity while providing a physically transparent interpretation of

curvature in terms of energy–pressure distributions. We clarify the non-local nature of the gravitational potential and the role of pressure as a source (not a direct bending agent).

Abstract

We present a consistent weak-field formulation of gravitational lensing within the Pressure–Potential Curvature (PPC) law of gravity. In this framework, gravity is sourced by a pressure-augmented energy density, expressed explicitly using gravitational pressure P_g. We derive the PPC stress–energy tensor, connect it to the Einstein field equations, and obtain the weak-field limit, deflection law, and angular lensing relations. The formulation reproduces the standard weak-field structure while providing a transparent interpretation of curvature in terms of energy–pressure distributions.

1. Introduction

Gravitational lensing—deflection of light by gravitating systems—is a cornerstone test of gravitational theory. In standard treatments, light follows null geodesics of a curved spacetime, and the deflection is governed by gradients of a gravitational potential in the weak-field limit.

The PPC law of gravity emphasizes the role of pressure in addition to energy density. Rather than treating curvature as purely geometric, PPC interprets the source of curvature as an energy–pressure structure. This paper develops a lensing formulation consistent with the weak-field limit and observationally relevant angular variables.

1. Introduction

Gravitational lensing is a fundamental test of gravity, arising from the curvature of spacetime. In PPC gravity, curvature is interpreted as arising from both energy density and gravitational pressure P_g, providing a physically motivated extension of standard formulations.

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2. PPC Stress–Energy Tensor

We define the PPC stress–energy tensor for a perfect fluid as:

$$T_{\mu\nu} = (E_d + P_g), u_\mu u_\nu + P_g, g_{\mu\nu}$$

where:

- E_d is energy density
- P_g is gravitational pressure
- u_μ is the four-velocity

If an equation of state is assumed:

$$P_g = \omega E_d$$

then:

$$T_{\mu\nu} = E_d \left[(1+\omega) u_\mu u_\nu + \omega g_{\mu\nu}\right]$$

3. PPC Field Equation

The curvature of spacetime is governed by the Einstein field equation:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Substituting the PPC tensor:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left[(E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu} \right]$$

This establishes that both energy density and gravitational pressure contribute directly to spacetime curvature.

4. Weak-Field Limit

In the weak-field approximation, the metric is expanded as:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

The PPC potential Φ_{PPC} satisfies:

$$\nabla^2 \Phi_{PPC} = 4\pi G \frac{(E_d + 3P_g)}{c^2}$$

This shows that pressure contributes with a factor of 3 in the effective gravitational source.

5. Light Bending in PPC Gravity

Light follows null geodesics. In the weak-field regime:

$$\frac{d^2 x_\perp}{dz^2} = -\nabla_\perp \Phi_{PPC}$$

The deflection angle is:

$$\alpha_{PPC} = \int \nabla_\perp \Phi_{PPC}, dl$$

Pressure influences light bending only through its contribution to the potential Φ_{PPC}, not as a direct force.

6. Angular Formulation

With impact parameter $b = D_L\theta$:

$$\alpha_{PPC}(\theta) = \frac{4G(E_d+3P_g)}{c^2 D_L \theta}$$

Lens equation:

$$\beta = \theta - \frac{D_{LS}}{D_S}\alpha_{PPC}(\theta)$$

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7. Einstein Radius

For $\beta = 0$:

$$\theta_E^{PPC} = \sqrt{\frac{4G(E_d+3P_g)}{c^2}\frac{D_{LS}}{D_L D_S}}$$

8. Physical Interpretation

In PPC gravity:

- Curvature is determined by $T_{\mu\nu}(E_d, P_g)$
- Pressure contributes directly to the gravitational source
- Light bending is governed by gradients of Φ_{PPC}

Thus:

> Pressure does not directly bend light, but modifies the curvature that governs light propagation.

9. Causal Structure

$$(E_d, P_g) \to T_{\mu\nu} \to G_{\mu\nu} \to \Phi_{PPC} \to \nabla\Phi_{PPC} \to \alpha_{PPC}$$

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10. Recovery of Standard Limit

For negligible pressure $(P_g \to 0)$:

$$\nabla^2\Phi_{PPC} \to 4\pi G E_d/c^2$$

and:

$$\alpha \to \frac{4GM}{bc^2}$$

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11. Limitations and Outlook

- Valid in weak-field approximation
- Assumes perfect (isotropic) fluid
- Requires full tensor extension for strong-field regime

12. Conclusion

We have developed a consistent gravitational lensing formulation in PPC gravity using gravitational pressure P_g. The theory reproduces known weak-field results while offering a new interpretation of curvature as arising from energy–pressure distributions.

General equation:

$$\alpha = \frac{4G}{c^4 b} \int (E_d + 3P_g)\, dV$$

- E_d: energy density
- P_g: gravitational pressure
- Integral → **total effective energy source**
- b: impact parameter

🔍 2. Special Case (constant ω)

Given:

$$P_g = \omega E_d$$

Then:

$$E_d + 3P_g = E_d(1 + 3\omega)$$

Substitute into integral:

$$\int (E_d + 3P_g)\, dV = (1 + 3\omega) \int E_d\, dV$$

But:

$$\int E_d\, dV = Mc^2$$

So:

$$\alpha = \tfrac{4G}{c^4 b}(1 + 3\omega)Mc^2$$

Cancel c^2:

$$\boxed{\alpha = \frac{4G(1 + 3\omega)M}{c^2 b}}$$

🔄 Angular form

Using:

$$b = D_L\theta$$

$$\boxed{\alpha_{PPC}(\theta) = \frac{4G(1+3\omega)M}{c^2 D_L\theta}}$$

Special case:

$$\alpha_{PPC} \propto (1 + 3\omega)$$

- Pressure modifies **effective gravitational mass**

🔥 4. Key PPC Insight

$$\boxed{\text{Effective source} = E_d + 3P_g}$$

$$\boxed{M_{eff} = \frac{1}{c^2} \int (E_d + 3P_g)\, dV}$$

✔ General form:

$$\alpha = \frac{4G}{c^4 b} \int (E_d + 3P_g)\, dV$$

✔ Special case:

$$\alpha_{PPC} = \frac{4G(1+3\omega)M}{c^2 b}$$

✔ Angular form:

$$\alpha_{PPC}(\theta) = \frac{4G(1+3\omega)M}{c^2 D_L \theta}$$

Keywords

PPC gravity, gravitational pressure, lensing, weak-field approximation, stress–energy tensor

$$\nabla^2 \Phi_{PPC} \propto (E_d + 3P_g)$$

$$\mathbf{g}_{PPC} = -\nabla \Phi_{PPC}$$

$$\alpha = \int \nabla_\perp \Phi_{PPC}\, dl$$

4. Gravitational field (what actually acts)

$$\mathbf{g}_{PPC} = -\nabla \Phi_{PPC}$$

- This is the **true bending agent**

5. Light deflection

$$\theta \sim \int \nabla_{\perp} \Phi_{PPC} \, dl$$

👉 Light responds to **potential gradient**, not directly to pressure

$$\boxed{\nabla P_g \;\Rightarrow\; \nabla(E_d + 3P_g) \;\Rightarrow\; \nabla^2\Phi_{PPC} \;\Rightarrow\; \nabla\Phi_{PPC} \;\Rightarrow\; \theta}$$

(2) Weak-field limit

$$\boxed{\nabla^2\Phi \propto (E_d + 3P_g)}$$

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(3) Role of pressure gradient

$$\boxed{\nabla P_g \text{ affects matter distribution, not curvature directly}}$$

> **Light bending in PPC gravity is governed solely by the gradient of the gravitational potential, which is generated by the combined energy–pressure distribution.**

- ✔ Pressure → **source of gravity**
- ✔ Potential Φ_{PPC} → **mediator**
- ✔ Gradient $\nabla\Phi_{PPC}$ → **actual bending agent**
- ✔ Deflection α → **observable**

$$\nabla \Phi_{PPC} \sim \text{effective field generated by } (E_d + 3P_g)$$

🔗 5. Correct logical chain

$$P_g \rightarrow (E_d + 3P_g) \rightarrow \Phi_{PPC} \rightarrow \nabla \Phi_{PPC} \rightarrow \alpha$$

$$\boxed{\nabla P_g \rightarrow \text{source variation} \rightarrow \Phi_{PPC} \rightarrow \nabla \Phi_{PPC} \rightarrow \theta}$$

✔ Deflection angle α

$$\boxed{\alpha = \text{physical bending angle of light due to gravity}}$$

- Caused by spacetime curvature
- Determined by:

$$\alpha = \int \nabla_{\perp} \Phi_{PPC} \, dl$$

👉 It is a **physical angle along the light path**

✔ Angular position θ

$$\boxed{\theta = \text{observed angular position of the image}}$$

- Measured by observer in the sky
- Related to impact parameter:

$$b = D_L \theta$$

👉 It is an **observational angle**

🔗 3. How they are connected

Through the **lens equation**:

$$\boxed{\beta = \theta - \frac{D_{LS}}{D_S}\alpha(\theta)}$$

$$\nabla\Phi_{PPC} \to \alpha \to \theta$$

👉 So:

$$\boxed{\alpha \text{ is cause, } \theta \text{ is observed effect}}$$

Field equation:

$$\nabla^2 \Phi_{PPC} \propto (E_d + 3P_g)$$

Light bending:

$$\alpha_{PPC} = \int \nabla_\perp \Phi_{PPC} \, dl$$

Interpretation:

$$\nabla \Phi_{PPC} \sim \text{effective pressure-gradient field}$$